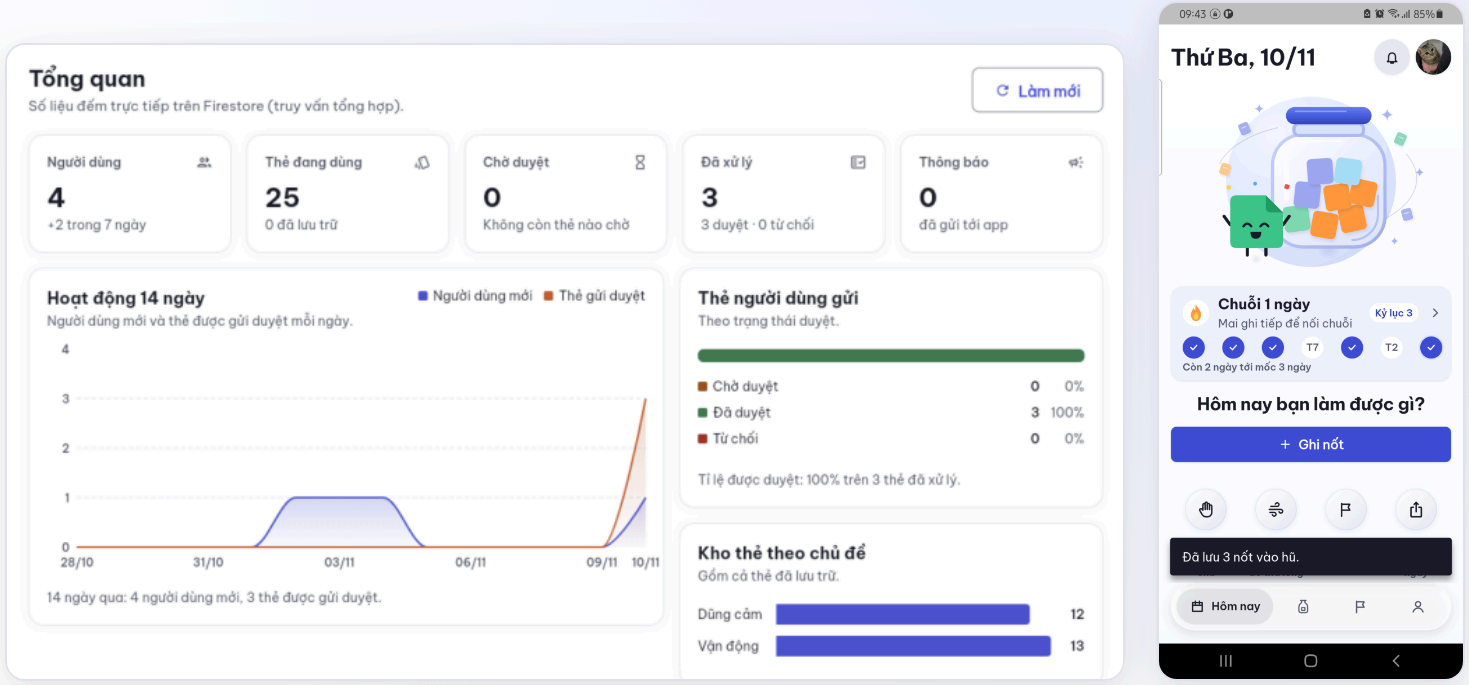
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# Nốt

## One positive note every day, now with staff, AI and a shared catalog

Đoàn Thị Kim Thúy · SE190992
Phùng Thế Quân · SE193519
Phan Đức Duy · SE193591
Phan Hữu Phước · SE161951



P1 PROBLEM 01 OVERVIEW

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# Nốt is a daily-wins app with a staff web admin, running on one Firebase project

The Lab 3 app became a two-client, three-role system: users log small wins on Android, staff curate the content they see.

### USERS

Final-year students who feel behind; staff and admins who run the content

### VALUE

Log a mood and one small win in under 30 s; get new challenge cards and news without updating the app

### PLATFORM

Flutter Android app · Flutter Web admin on Hosting · Firestore, Rules, App Check, Gemini

### WHAT THE FINAL ADDED ON TOP OF LAB 3

- Roles `user` / `staff` / `admin` and a web admin with 5 screens
- Cross-role flow: a user shares a card, staff approve it, the app updates in real time
- Nốt AI (Gemini via Firebase AI Logic): tell the day, get sorted notes to confirm
- Inbox for announcements, first-run onboarding, weekly card PDF saved on the device




Live: prm393-not-lab3.web.app (app) · prm393-not-lab3.web.app/admin/ (web admin) · repo FU-PRM-Projects/Lab-03

# Minh ends most evenings remembering what he did not finish

Lab 2 persona and the habit Nốt was designed around; Lab 3 shipped it for one person, alone.

### PERSONA

**Minh, 22**, final-year student, worn out. Ends the evening looking at the list of unfinished tasks instead of what went well.

### SUCCESS CRITERION (LAB 2)

Log a mood and one small win in **under 30 seconds** after tapping the evening reminder.

### HOW NỐT WORKED AFTER LAB 3

- One user, own data only (notes, habits, groups)
- Challenge cards: a fixed seed list in Firestore
- Own cards stayed private to their author

### WHO ELSE IS INVOLVED NOW

- Staff who write and curate challenge cards
- An admin who decides who is staff
- Gemini, to cut typing in the evening




Weekly card, Lab 3 flow kept in the Final

Source: Lab 2 persona and success criterion (README, docs/design-traceability.md).

P1 PROBLEM 03 PROBLEM

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# Three gaps kept Nốt from being a system: no curator, no channel, slow evenings

Each pain point is answered directly on the next slide.

### Static challenge content

CAUSE

Cards were a seed list; changing one needed a developer and a script

IMPACT

**Users run out of fresh cards; good ideas from users stay private**

### No way to reach users

CAUSE

No staff role, no announcements, push only from the Console

IMPACT

**Nobody can tell users about new cards or events inside the app**

### Logging takes effort at night

CAUSE

Each note is typed and classified by hand (win, self-control, journal)

IMPACT

**The 30 s goal slips when the day had several moments to record**

### Lab 3 flow left unfinished

CAUSE

The weekly card PDF went to Cloud Storage, which needs the paid plan

IMPACT

**Export never worked on the real project**

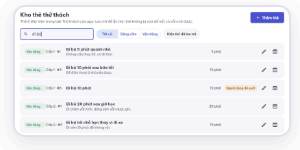
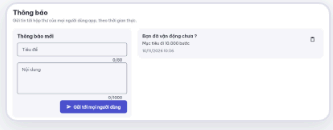
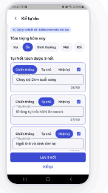


Evidence: docs/gap-review.md (Lab 2 / Lab 3 gaps before the Final). No user survey was run; impacts are observed in the Lab 3 build.

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# One Firebase system: a staff web admin, a review flow, and Gemini that sorts the day

Every pain point maps to a feature that is live and demonstrated.

| PAIN POINT | SOLUTION | EXPECTED VALUE | LIVE |
|---|---|---|---|
| Static content | Web admin catalog + users share cards, staff approve | Catalog grows without a release |  |
| No channel | Announcements + personal inbox (Firestore, real time) | News lands in the app in the same minute |  |
| Slow evenings | Nốt AI: tell the day once, confirm sorted notes | Several notes from one paragraph |  |
| Broken PDF | Build and save the PDF on the device, metadata in Firestore | Works on the free plan, offline |  |

### IN SCOPE (MVP)

- WF-01 Log the day with Nốt AI
- WF-02 Publish a user card to the catalog
- WF-03 Announce news to every user
- Onboarding, weekly PDF, roles, App Check

### OUT OF SCOPE

- Cloud Functions, Cloud Storage (brief + paid plan)
- AI level 3 "Hỏi Nốt" (bonus, not built)
- iOS build

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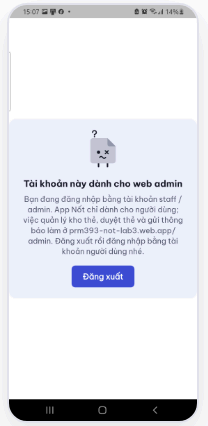
# Three human roles and two external services, enforced by Security Rules

The role lives in users/{uid}.role; only an admin changes it, and never their own.

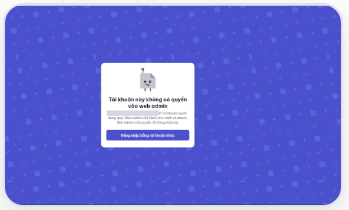
| ACTOR | CLIENT | WHO | MAIN PERMISSIONS |
|---|---|---|---|
| **User** | Android app | Student using Nốt every evening | Own notes and data; share own cards (pending only); read catalog, announcements, own inbox |
| **Staff** | Web admin | Content curator | Create, edit, archive catalog cards; approve / reject with reason; post announcements |
| **Admin** | Web admin | Team lead | Everything staff can do + change other users' roles |
| **Gemini** | External | Firebase AI Logic, Gemini Developer API | Receives the scrubbed paragraph only; returns JSON suggestions |
| **Firebase** | Platform | Auth, Firestore, App Check, RC, FCM… | Rules decide every read and write; no custom server |




Admin changes roles (own role locked)




Staff/admin on the phone



User on the web

First admin: set in the Firebase Console (Thị Kim Thúy Đoàn). Rules tests R-01…R-30 cover every role and collection (BR-05: R-26).

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# Three core workflows carry the demo; everything else supports them

Each workflow has an actor, a trigger and a business result, not a list of buttons.

WF-01

## Log the day with Nốt AI

| | |
|---|---|
| ACTOR | User |
| TRIGGER | Evening, user taps Ghi nốt → Kể tự do |
| RESULT | **Notes saved after the user confirms** |
| RULES | BR-01 · consent, Gemini JSON, edit, fallback |
| SOLVES | Slow evenings |

WF-02

## Publish a user card to the shared catalog

| | |
|---|---|
| ACTOR | User + Staff |
| TRIGGER | User shares an own card |
| RESULT | **Card in the catalog, app updates, inbox message** |
| RULES | BR-02 submit · BR-03 approve / reject + reason |
| SOLVES | Static content |

WF-03

## Announce news to every user

| | |
|---|---|
| ACTOR | Staff |
| TRIGGER | Staff has news for everyone |
| RESULT | **Announcement published; every inbox shows it** |
| RULES | BR-04 title 1–80, body 1–1000, staff only |
| SOLVES | No channel |

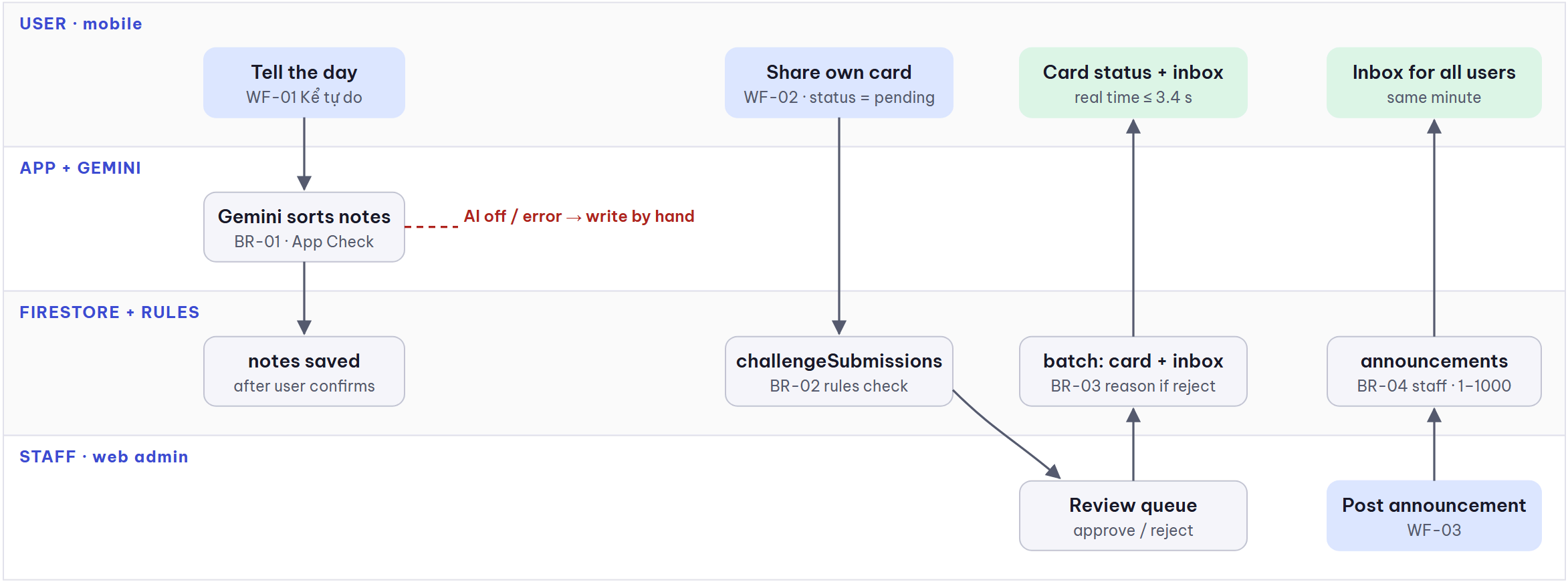


Supporting features: onboarding & starter plan · weekly card PDF · streaks & journey · self-control & breathing · groups · Discord link · dashboard · roles.

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# One swimlane view: who acts, where the data lands, where it branches

The three workflows share one Firestore; rules are the decision points.

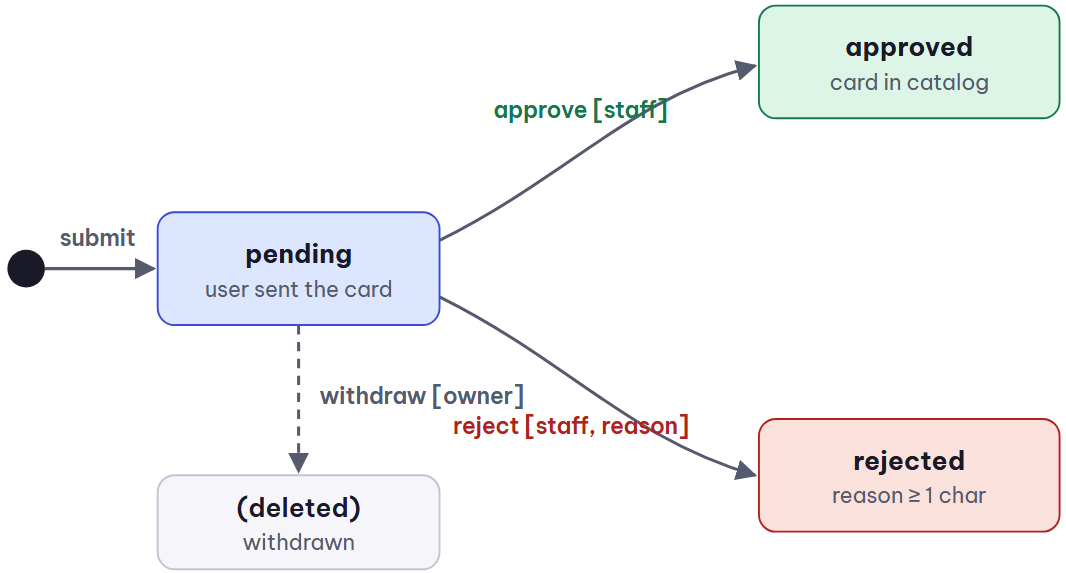



BR-01 AI only when enabled, consented and App Check passes · BR-02 user creates pending only · BR-03 reject needs a reason · BR-04 announcements: staff only, title 1–80, body 1–1000 · BR-05 only admins change roles, never their own.

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# A shared card moves pending → approved or rejected, and nowhere else

Entity: challengeSubmissions/{id}. Every transition and every forbidden one has a rules test.




**Forbidden (rules deny)**

user → approved
approved → pending
reject without reason
withdraw after review

```
109                     && request.resource.data.createdAt == request.time;
110         // Only pending -> approved | rejected; a rejection needs a reason.
111         allow update: if isStaff()
112                       && resource.data.status == 'pending'
113                       && request.resource.data.status in ['approved', 'rejected']
114                       && request.resource.data.diff(resource.data).affectedKeys()
115                            .hasOnly(['status', 'reason', 'reviewedAt', 'reviewedBy'
116                       && request.resource.data.reviewedBy == request.auth.uid
117                       && request.resource.data.reviewedAt == request.time
118                       && shortText(request.resource.data.get('reason', ''), 0, 200)
119                       && (request.resource.data.status == 'approved'
```

firestore.rules · the guard on every review (BR-03)

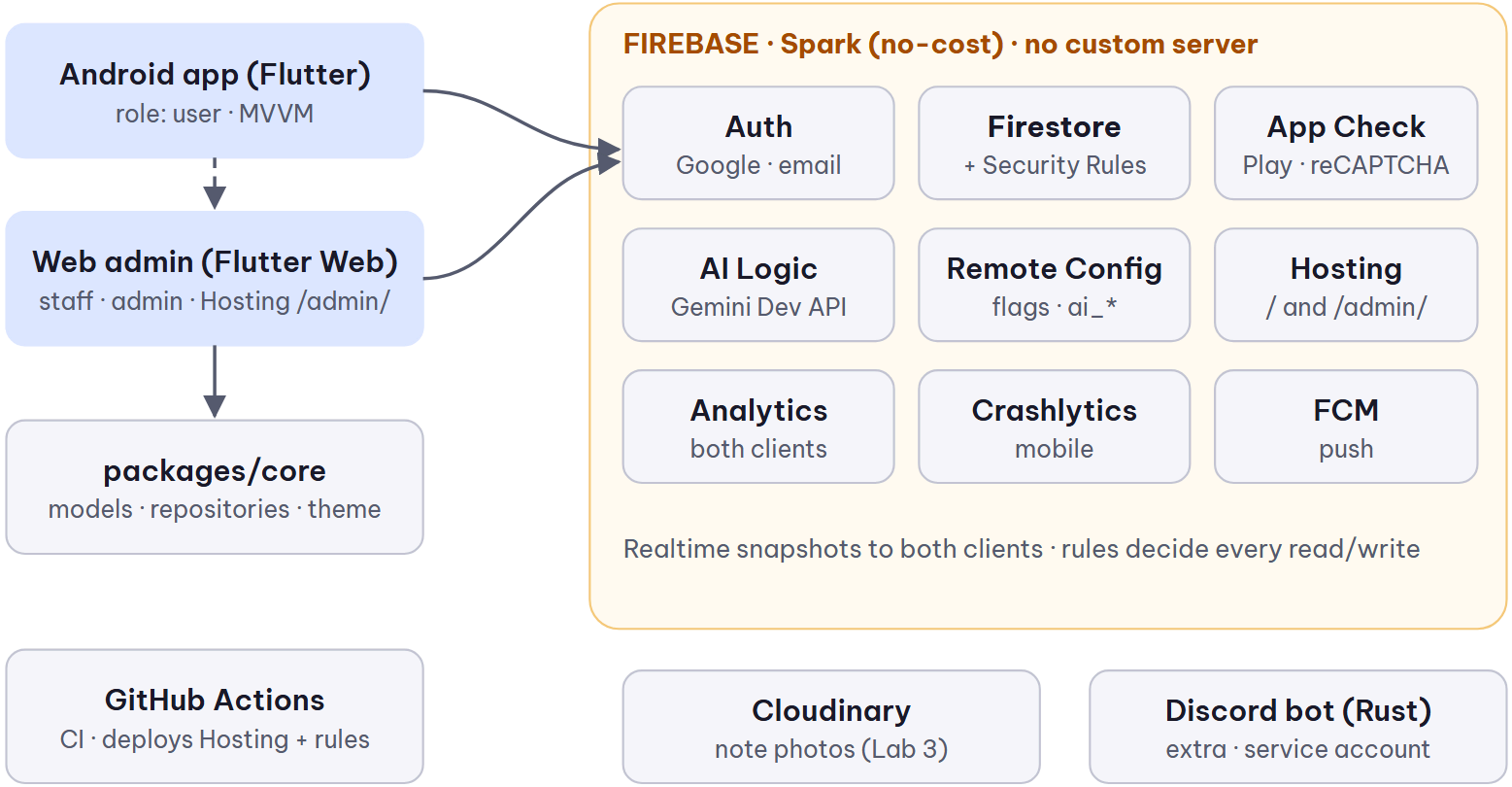


Tests: R-20 create pending only · R-21 read own or staff · R-22 staff review, reason on reject · R-23 withdraw while pending. 30/30 rules tests pass.

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# Two Flutter clients, one shared package, Firebase as the only backend

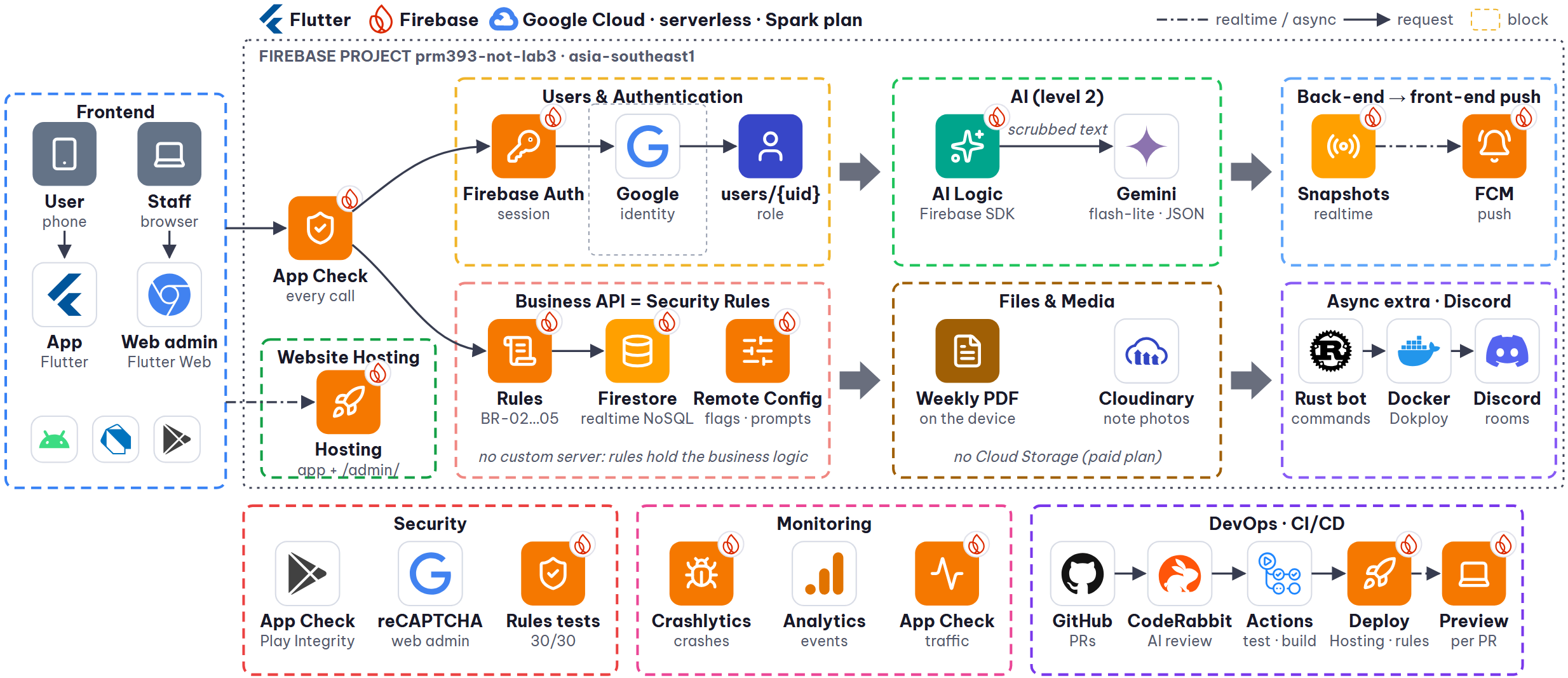
Decision: no custom server. Security Rules + App Check are the backend's access layer.




Deploy: GitHub Actions builds both clients, deploys Hosting (/ and /admin/) and Firestore rules + indexes on merge to main.

# Serverless on Firebase: Security Rules are the backend

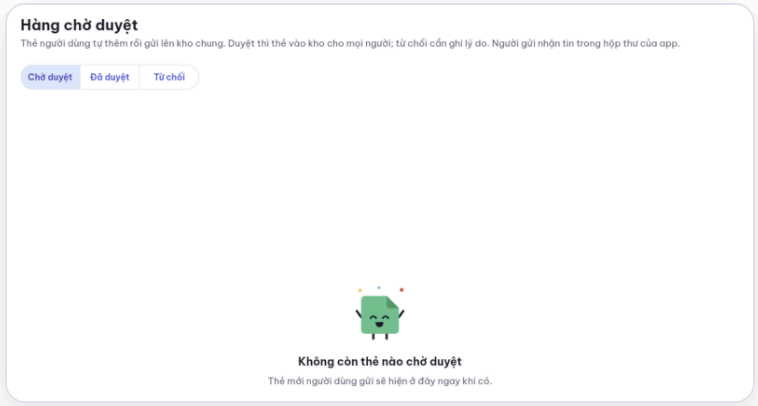




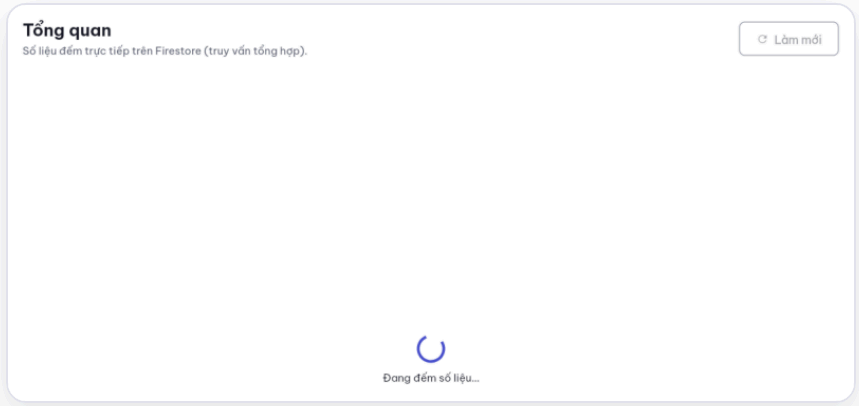
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# One design system across phone and web, with every UI state designed

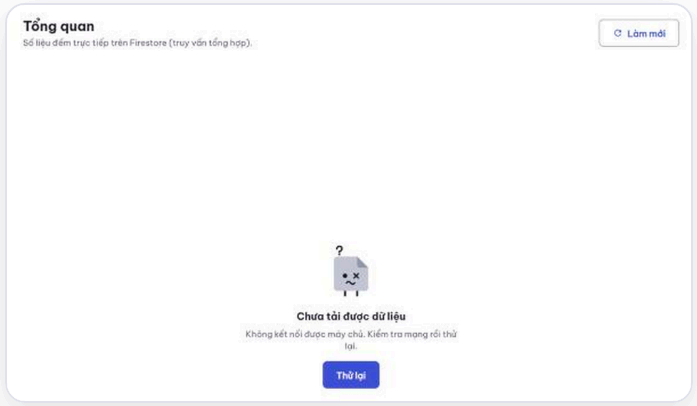
Lab 2 tokens live in packages/core; the web admin reuses them in a wide layout.



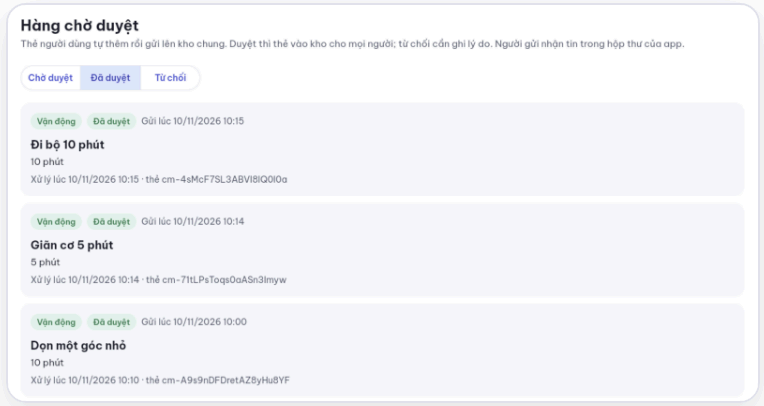

Empty · Hàng chờ duyệt



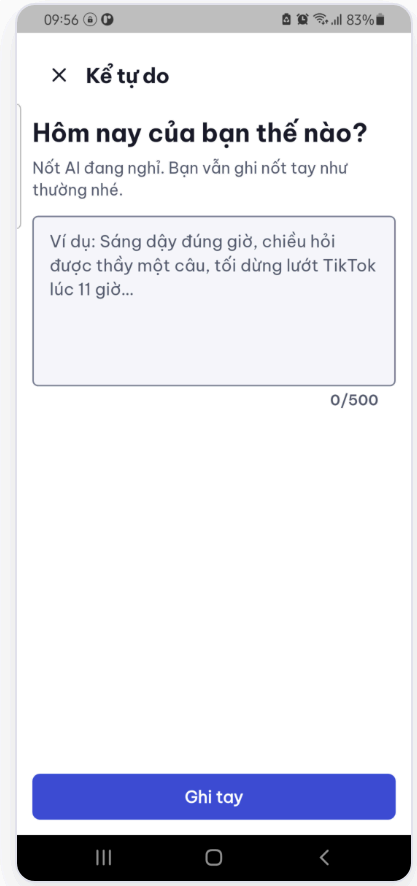

Loading · Tổng quan




Error · server unreachable, Thử lại



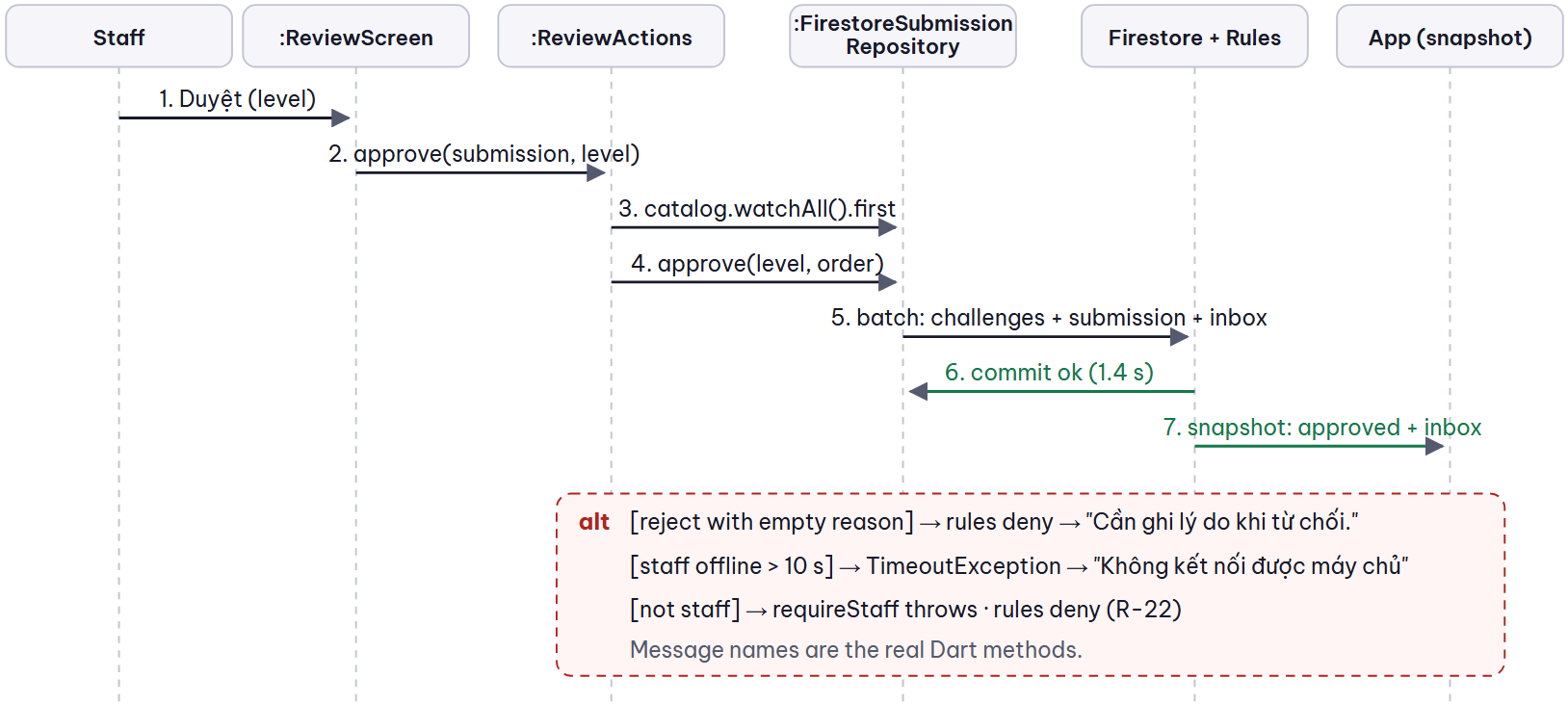

Success · approved cards




App · AI off state

States: empty · loading · error (within 15 s, Thử lại) · success · wrong role rejected on web and mobile.

# Approving a card is one batch write; rules guard it, snapshots carry it back




```
        })
        ..update(_col.doc(submission.id), <String, Object?>{
          'status': SubmissionStatus.approved.id,
          'reason': '',
          'challengeId': cardId,
          'reviewedBy': reviewerUid,
          'reviewedAt': FieldValue.serverTimestamp(),
        })
        ..set(_inbox(submission.uid).doc(), <String, Object?>{
          'kind': InboxKind.submissionApproved.id,
          'title': 'Thẻ của bạn đã vào kho chung',
          'body': _clip(
            '"${submission.title}" đã được duyệt. Mọi người giờ có thể nhận '
            'thẻ này ở tab Thử thách.',
            500,
          ),
          'submissionId': submission.id,
          'read': false,
          'createdAt': FieldValue.serverTimestamp(),
        });
      await batch.commit();
    });
```

FirestoreSubmissionRepository.approve: one batch

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# MVVM in both clients: screens never touch Firebase directly

Business rules sit in the repository + Security Rules, not in widgets.

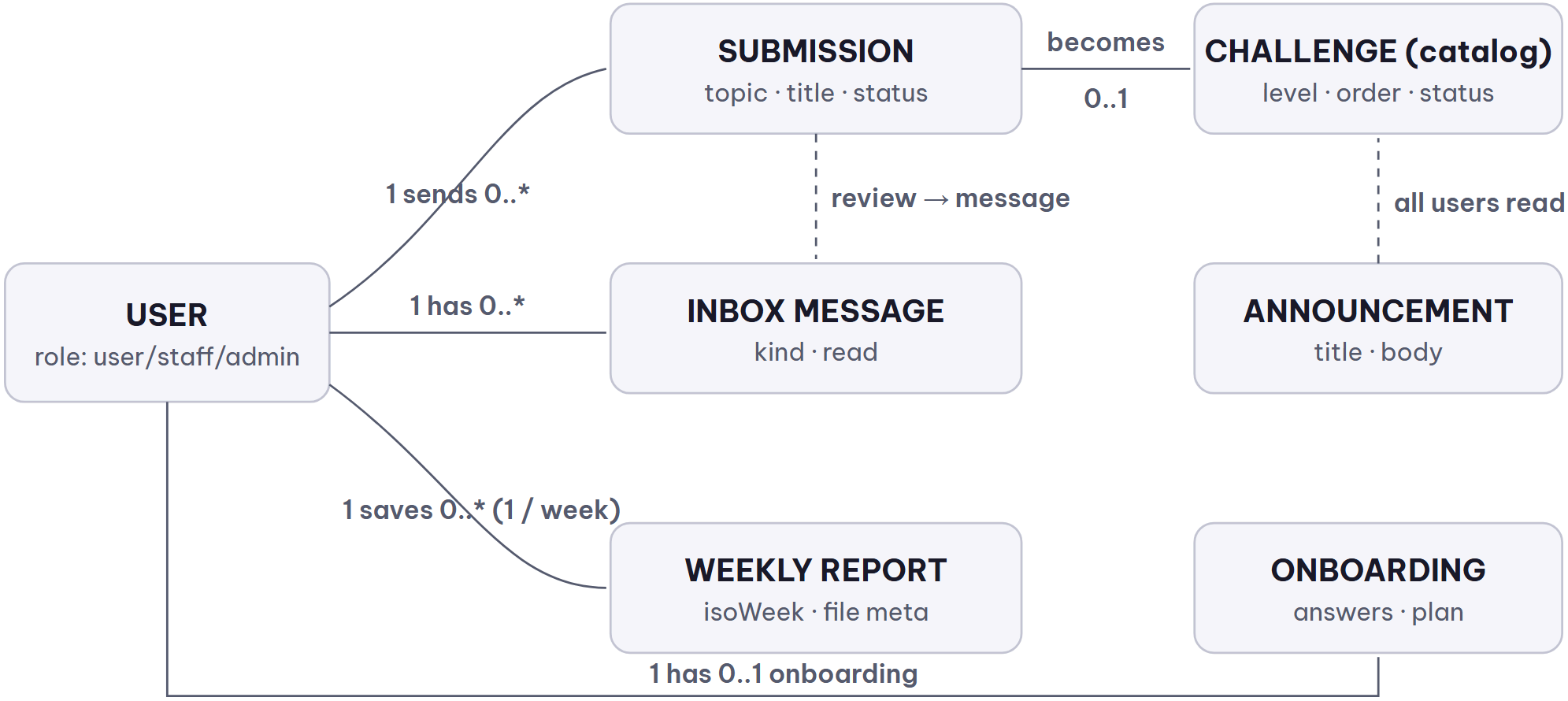
| LAYER | WF-01 NỐT AI | WF-02 SHARE & REVIEW | WF-03 ANNOUNCE |
|---|---|---|---|
| **Screen** | `FreeTellScreen` | `ChallengesScreen` · `ReviewScreen` | `AnnouncementsScreen` · `NotificationInboxSheet` |
| **ViewModel** | `FreeTellViewModel` | `CommunityActions` · `ReviewActions` | `AnnouncementActions` · `communityInboxProvider` |
| **Service** | `AiService` | `SubmissionRepository` · `CatalogRepository` | `AnnouncementRepository` · `InboxRepository` |
| **Implementation** | `FirebaseAiService` | `FirestoreSubmissionRepository` | `FirestoreAnnouncementRepository` |
| **Model** | `AiSortResult` · `AiNoteSuggestion` | `ChallengeSubmission` · `CatalogCard` | `Announcement` · `InboxMessage` |



main.dart wires each provider to its Firebase implementation; tests swap in fakes (303 app · 14 web admin · 5 core tests).

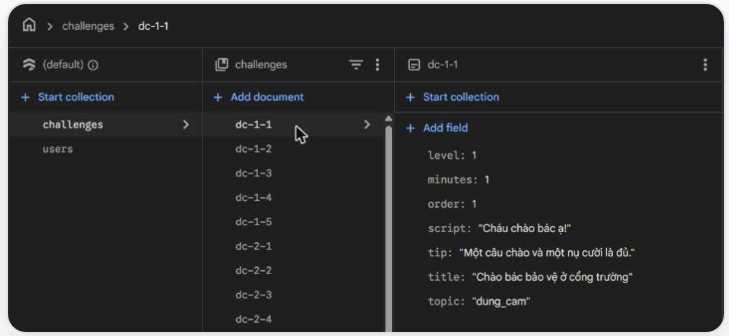
# The Final added six entities around the user

Lab 3 entities (notes, habits, groups, progress) stay private to their owner.



# Firestore collections, with constraints written as Security Rules

| COLLECTION | KEY FIELDS | CONSTRAINTS (RULES) |
|---|---|---|
| users/{uid} | role, displayName, email | self-create as user; only admin changes role, never own |
| challenges/{id} | topic, level 1–10, order, title ≤ 80, status | staff only; updatedBy = auth.uid; archive, no delete |
| challengeSubmissions/{id} | uid, topic, title, minutes, status, reason | create pending; staff review; reason on reject |
| users/{uid}/inbox/{id} | kind, title, body, read | staff create; owner reads, may only flip read |
| announcements/{id} | title ≤ 80, body ≤ 1000, authorUid | staff create/delete; every signed-in user reads |
| users/{uid}/reports/{week} | isoWeek, fileName, sizeBytes, noteCount | owner only; id = week; savedAt = server time |
| users/{uid}/private/onboarding | answers, profileLabel, planSource | owner only; enums and sizes checked |




Firestore Console · challenges

```
127   // the app inbox.
128   match /announcements/{id} {
129     allow read: if signedIn();
130     allow create: if isStaff()
131                   && request.resource.data.keys().hasOnly(['title', 'body',
132                   && request.resource.data.keys().hasAll(['title', 'body',
133                   && shortText(request.resource.data.title, 1, 80)
134                   && shortText(request.resource.data.body, 1, 1000)
135                   && request.resource.data.authorUid == request.auth.uid
136                   && request.resource.data.createdAt == request.time;
137     allow delete: if isStaff();
```

Rules for announcements (BR-04)

Indexes: submissions (uid, createdAt↓), (status, createdAt↓) · notes (groupId, createdAt↓) · full table: docs/data-model.md

# Each workflow traces from screen to rule to test

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| WORKFLOW | SCREENS | CLASS / SERVICE | COLLECTION | TESTS |
|---|---|---|---|---|
| **WF-01** Nốt AI | S03b Kể tự do | FreeTellViewModel · AiService | notes | BR-01 · free_tell_flow_test · ARTEMIS T18 (2/3), T19 (3/3) |
| **WF-02** Share & review | S07 · Review queue · Inbox | CommunityActions · ReviewActions | challengeSubmissions · challenges · inbox | BR-02/03 · R-20…R-24 · ARTEMIS T20 (3/3) · D-1…D-3 |
| **WF-03** Announce news | Announcements · Inbox | AnnouncementActions | announcements · inbox | BR-04 · R-25 · notification_inbox_test · web checklist #9–11 |
| Weekly PDF | S11b Thẻ tuần | WeeklyReportViewModel | reports | R-28 · weekly_report_test · ARTEMIS T14 (3/3) |
| Onboarding | S13a–g | OnboardingViewModel | private/onboarding | R-27 · onboarding_flow_test · ARTEMIS T17 (3/3) |
| Roles | Sign-in · Users | sessionProvider · UserActions | users | BR-05 · R-01, R-26, R-29 · session_test · auth_flow_test |



Full matrix: docs/design-traceability.md · rules: firebase/tests/firestore.rules.test.js · E2E: e2e_test/

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# The demo is one evening: Minh logs his day, shares a card, staff publish it and announce

The result of each workflow is the input of the next.

**01 · USER**
Kể tự do: one paragraph → 3 notes saved

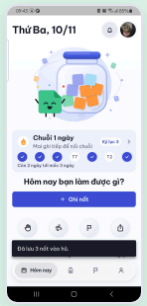
→

**02 · USER → STAFF**
Share "Đi bộ 10 phút" → staff approves → app shows it

→

**03 · STAFF → USERS**
"Bạn đã vận động chưa?" → every inbox

## RESULT WF-01



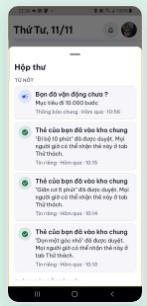

Jar 11 → 14 notes, streak starts

## RESULT WF-02




Card "Người dùng đề xuất" in catalog; inbox message

## RESULT WF-03




Announcement in app inbox at 10:56, same minute

## EXCEPTION PATHS SHOWN LIVE

- Remote Config `ai_enabled = false` → "Nốt AI đang nghỉ", user writes by hand
- Reject without reason → blocked (BR-03) · user account on the web, staff account on the phone → rejected (BR-05)

Prepared: user account on a Galaxy A70 (release build, Play Integrity) · staff/admin account on the web admin · ai_enabled ready to toggle in Remote Config · empty review queue · backup video.

# WF-01 Log the day with Nốt AI: one paragraph becomes three editable notes

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### TRIGGER · PRECONDITION

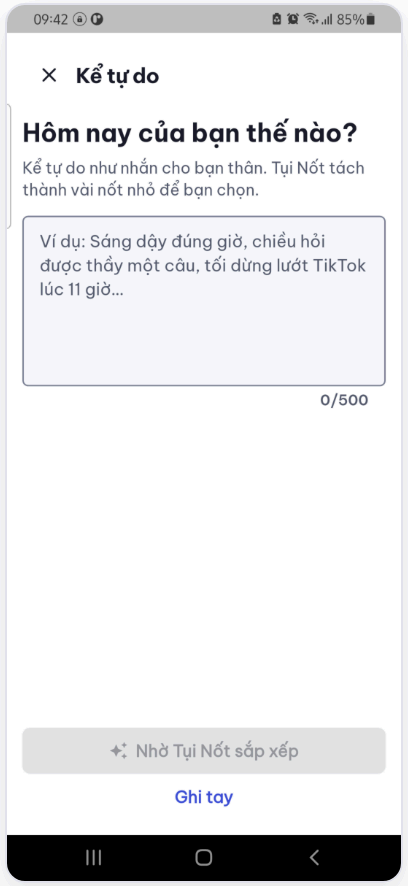
Evening, Ghi nốt → Kể tự do · role `user`, AI on (BR-01)

### STEPS → RESULT

- Consent sheet on first use
- Scrubbed paragraph → Gemini → JSON notes + mood
- Labelled "Gợi ý từ Nốt AI", editable
- **Result:** chosen notes saved, jar and streak update

### EXCEPTION

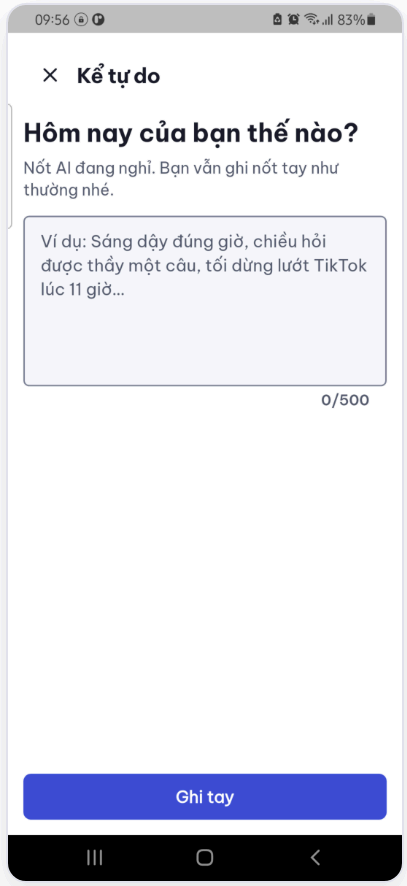
BR-01 fails (AI off, App Check, network) → "Ghi tay", paragraph kept




Consent




Suggestions




ai_enabled = false

Evidence: Galaxy A70, real Gemini (~2 s) · ARTEMIS T18 2/3, T19 (AI off) 3/3 · AI level 2: JSON schema → AiSortResult.

P3 DEMO 19 WF-02 · USER + STAFF

# WF-02 Publish a card: staff approve on the web, the phone updates within seconds

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### TRIGGER · STATE

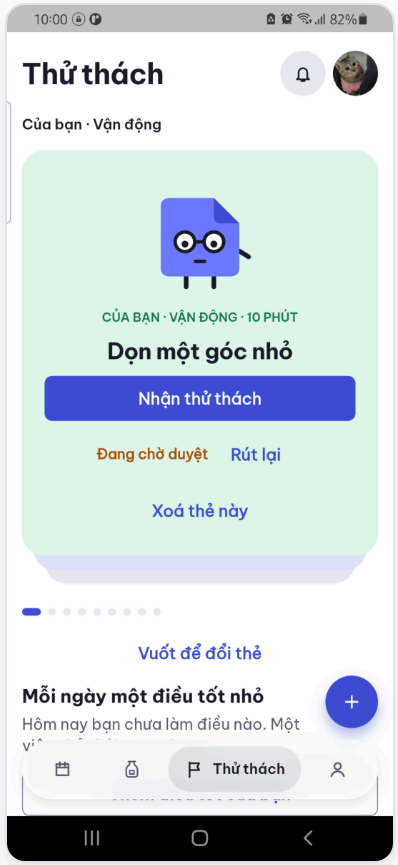
User shares own card · **pending** → **approved** / rejected

### STEPS

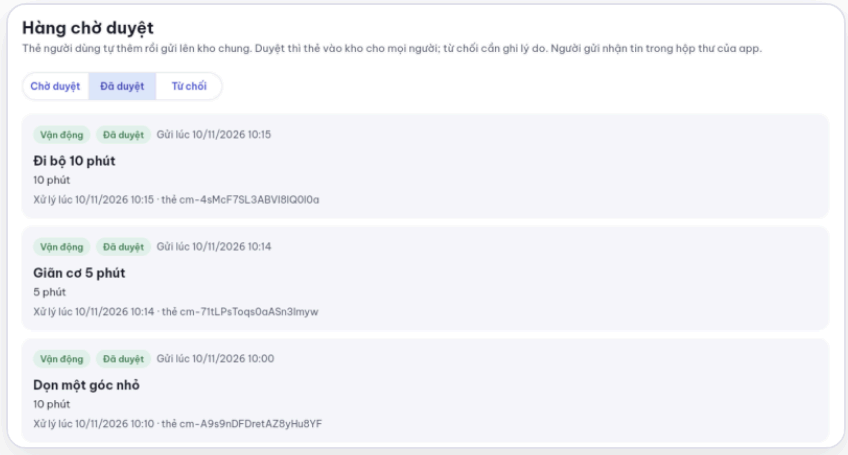
- User shares → pending
- Staff: Duyệt on the web
- Batch: card + status + inbox
- App: "Đã vào kho chung ✓"

### EXCEPTION

BR-03 reason required · BR-02 user cannot approve




Đang chờ duyệt




Web: approved




App: inbox

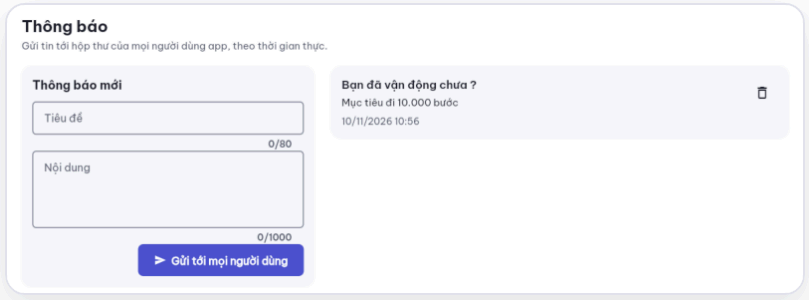
Evidence: write 1.4 s after Duyệt · phone ≤ 3.4 s (D-2, D-3), 22 s when idle (D-1) · ARTEMIS T20 3/3 · docs/cross-role-flow.md

P3 DEMO 20 WF-03 · STAFF

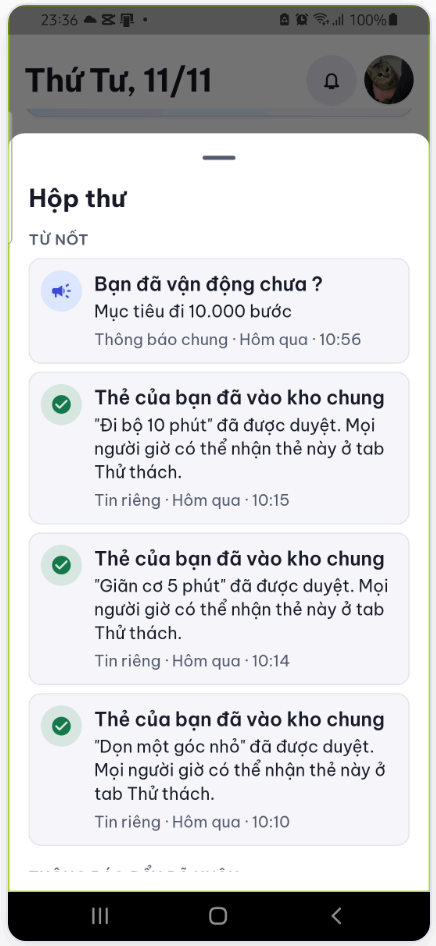
# WF-03 Announce news to every user: staff post once, every inbox shows it in the same minute




1 · Compose (list still empty)




2 · Sent at 10:56, form cleared




3 · App inbox, 10:56

**TRIGGER · PRECONDITION**

Staff has news for every user · signed in as staff or admin

**DECISION (BR-04)**

Title 1–80 · body 1–1000 · authorUid = self, in the form and in the rules

**RESULT**

Announcement published; every user's inbox shows it in real time

**EXCEPTION**

Empty or too long → form error, nothing saved · user account → rejected at sign-in, rules deny (R-25)

Evidence: sent at 10:56, in the app inbox at 10:56 · rules R-25 (staff only, authorUid = self, title 1–80, body 1–1000) · web checklist #9–11.

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# Rules, code and the deployed system are each tested, with evidence in the repo

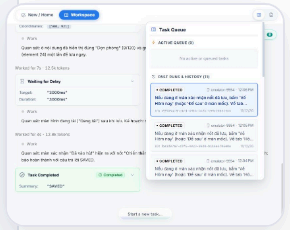
**30/30**
Security Rules tests

**303**
Flutter app tests

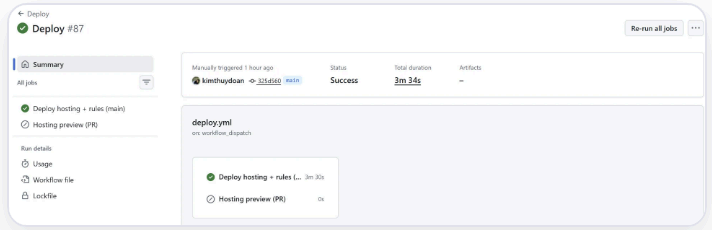
**14 + 5**
web admin + core tests

**14/15**
ARTEMIS Final runs, real phone

| LEVEL | EVIDENCE | RESULT |
|---|---|---|
| **Rules** | Emulator, allowed + denied per role per collection (BR-02…BR-05) | 30 / 30 |
| **Unit / widget** | ViewModels, repositories, screens at 360 dp | all pass |
| **Cross-role** | 3 runs on the deployed system, timed | 3 / 3 |
| **Web checklist** | Actions, rejection, empty, loading, error, live role change | 16 / 16 |
| **Real device** | Onboarding, Nốt AI, PDF, AI off, journal draft, role rejection on Galaxy A70 | pass |
| **E2E (ARTEMIS)** | Lab 3: 41/42 · Final T14, T17–T20 on Galaxy A70: 14/15 · T15/T16 not run | 14 / 15 |



ARTEMIS: agent reports SAVED




GitHub Actions: deploy on main

CI on every PR: format, analyze, all test suites, rules in the Emulator, web build, Android build, Hosting preview · details: docs/test-report.md.

P4 PROOF 22 ACHIEVEMENTS

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# Every brief requirement maps to a live feature and a piece of evidence

| BRIEF REQUIREMENT | IMPLEMENTED | EVIDENCE |
|---|---|---|
| Web admin: 4 screens + announcement form, deployed | Dashboard, catalog, review, announcements, users | `evidence/web-admin/` (16 items) |
| Role-based rules, deny by default, tested | user / staff / admin, R-01…R-30 | `evidence/firestore/rules-tests.txt` |
| Cross-role flow, real time, documented | Share card → approve → app + inbox | `docs/cross-role-flow.md`, `evidence/cross-role/` |
| AI level 2 with RC, App Check, fallback | Kể tự do on Gemini, `ai_*` flags incl. prompts, enforced App Check | `docs/ai-feature.md`, ARTEMIS T18/T19 |
| App Check on Firestore and AI Logic | **Enforced** on both · Web: reCAPTCHA Enterprise; Android: Play Integrity | `evidence/app-check/` |
| Lab 3 PDF saved & shared on device | pdf + share_plus, metadata in Firestore | 15.4 KB file pulled from the phone |
| Mobile data on Firestore with inbox | Announcements + personal inbox, real time | web checklist #11 |

**Not met:** ARTEMIS T15/T16 (email sign-in) not run → slide 23

P4 PROOF 23 LIMITATIONS & NEXT STEPS

# What is not done yet, and what it costs

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| LIMITATION | IMPACT | PRIORITY | NEXT STEP |
|---|---|---|---|
| ARTEMIS T15/T16 (email sign-in) not run | Sign-out / sign-in by email has widget tests only, no E2E on a device | Medium | Create a test-only account, run 3× each |
| Names can reach Gemini (CR-F2) | Email and phone are scrubbed, but a friend's name is sent as typed | Medium | Consent sheet says so; add on-device name masking |
| Exploratory session X01: 10 UI issues not rechecked | Possible small layout bugs on rarely used screens | Low | Re-run X01 on the Final build, fix or close each |
| D-1 took 22 s on an idle phone | Real time not guaranteed after long idle | Low | Measure more runs; re-listen on resume |
| Lab 2 flow A3 (practice chat) dropped | Overlaps the Discord rooms; one designed flow not built | Low | Reason in docs/gap-review.md §1.1 |

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# Nốt is now a working multi-role system, and we can show every claim

### 1 · Problem solved

Content is curated by staff and grows from users' own cards; news reaches every user in the same minute.

### 2 · Working product

WF-01, WF-02, WF-03 run live on Android and the deployed web admin, on the free plan.

### 3 · Verified

30/30 rules tests, 300+ code tests in CI, 14/15 ARTEMIS runs on a real phone, 3 timed cross-role runs, App Check enforced.

prm393-not-lab3.web.app
prm393-not-lab3.web.app/admin/
github.com/FU-PRM-Projects/Lab-03
Thank you · Questions?

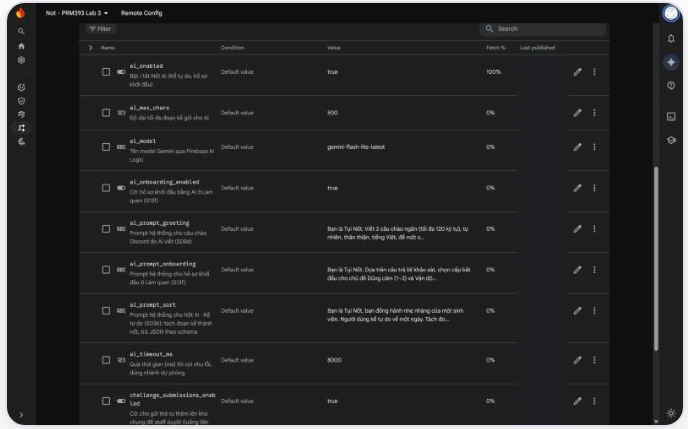
APPENDIX A1 · AI EXAMPLES

# Three real Gemini inputs and outputs, and one handled failure

| INPUT (KỂ TỰ DO) | GEMINI OUTPUT |
|---|---|
| Ran 2 km, ate a snack I meant to skip, calmed down without blaming myself | chien_thang "Chạy bộ 2km quanh hồ" · nhat_ky "Lỡ ăn thêm gói snack..." · tu_chu "Không tự trách và bình tĩnh lại" · mood binh_thuong |
| Asked a stranger in the library where to charge my phone; wrote a few lines about the rainy afternoon | chien_thang "Lấy hết can đảm hỏi mượn chỗ sạc pin" · nhat_ky "Ghi lại vài dòng cảm xúc buổi chiều mưa" · mood vui |
| Tired for days, only tidied my desk today | chien_thang "Mình đã dọn dẹp được cái bàn học" · 2× nhat_ky · mood met · needs_support false |
| **Handled failure** | 404 model retired / 401 missing App Check → `ai_error` logged, paragraph kept, "Ghi tay" offered; fixed by Remote Config model name |

```
29
30  GenerativeModel _model(String instructions, Schema schema) =>
31      FirebaseAI.googleAI().generativeModel(
32        model: config.model,
33        systemInstruction: Content.system(instructions),
34        generationConfig: GenerationConfig(
35          responseMimeType: 'application/json',
36          responseSchema: schema,
37          temperature: 0.4,
38        ),
39      );
```

JSON schema forces the answer shape

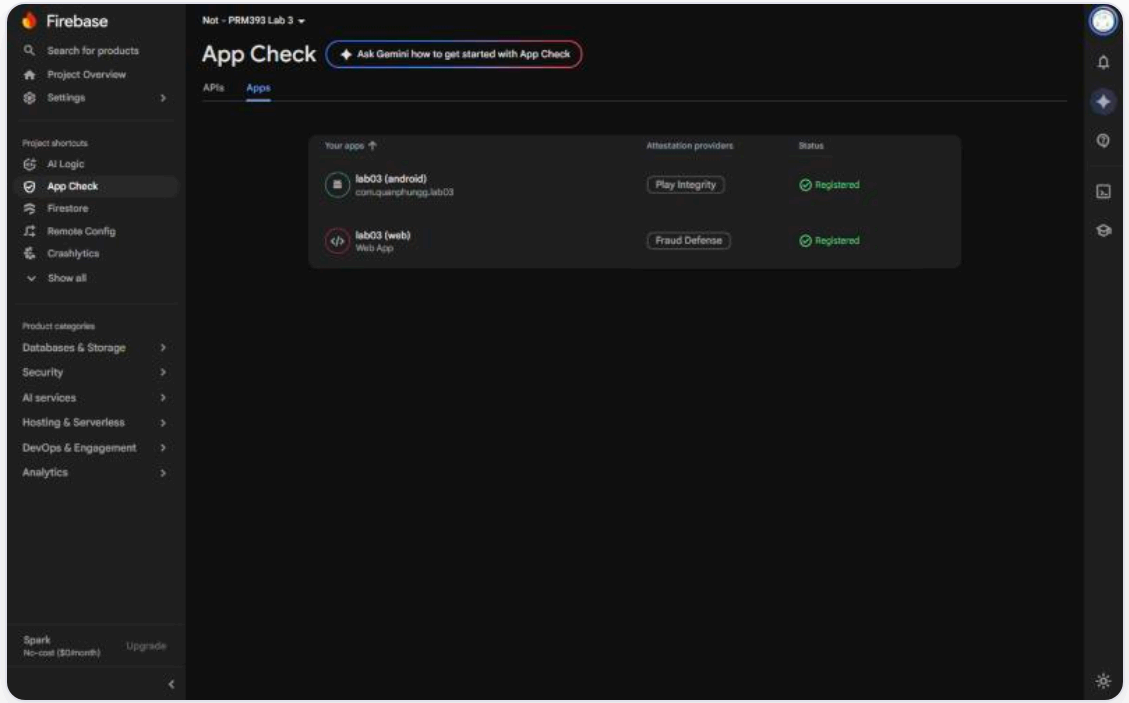


Remote Config: model, prompts, on/off

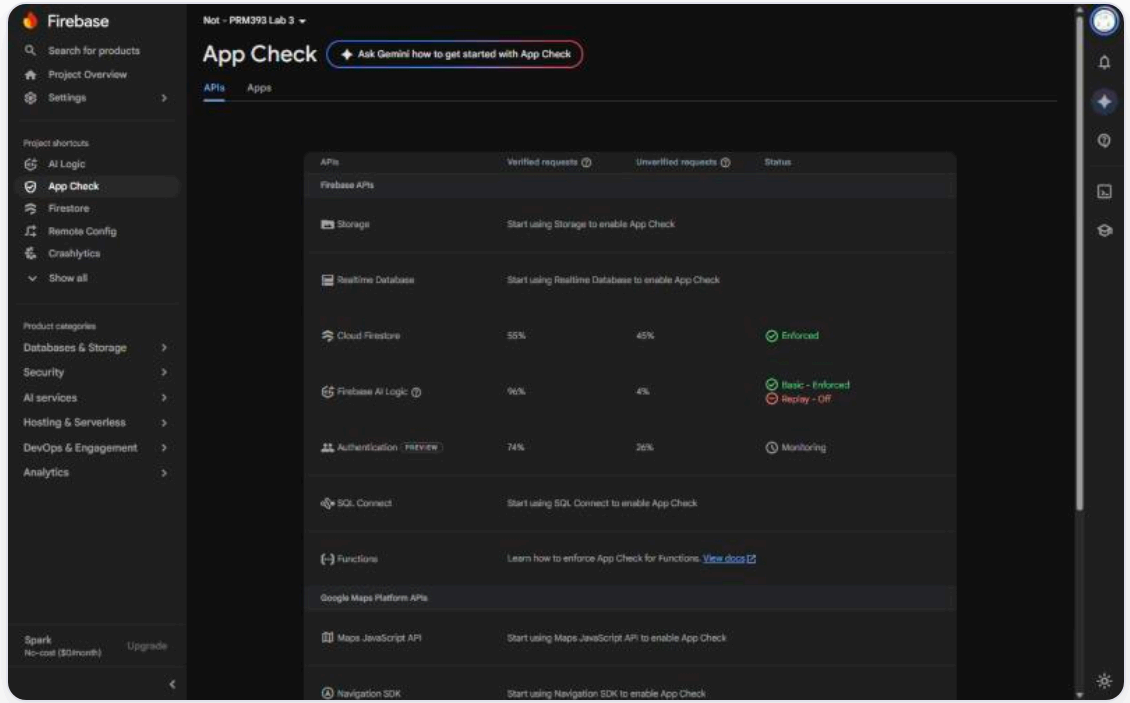
Inputs are scrubbed (email, phone) and capped at 500 characters; no names, emails or photos are sent.

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# Both apps are registered, and Firestore and AI Logic are enforced




Apps: lab03 (android) Play Integrity · lab03 (web) Fraud Defense




Cloud Firestore: Enforced · AI Logic: Enforced

Web: reCAPTCHA Enterprise (Fraud Defense) · Android: Play Integrity (debug builds need a registered debug token) · enforced 12/11/2026 · evidence/app-check/.

APPENDIX A3 · LESSONS LEARNED

# Running the system for real found bugs the tests did not

Five bugs surfaced only against the real project or a real device; all are fixed, with regression tests.

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### APPROVE TIMED OUT

Riverpod paused the unwatched catalog provider; approval waited 10 s and failed. Found while timing D-1. Fixed: read the repository once (PR #70).

### DASHBOARD SPUN OFFLINE

`count()` waits for the server forever. Found while shooting the error state. Fixed: 15 s timeout → error state (PR #71).

### ERROR STATES 3 MIN LATE

Riverpod 3 retried failed providers 10× while showing loading. Found on the deployed web admin. Fixed: `retry: noRetry` in both apps (PR #77).

### ROLE READ ONLY ONCE

A demoted staff kept the admin UI until reload (CR-F1). Fixed: the session follows the user's role document live (PR #75).

### GEMINI MODEL RETIRED

`gemini-2.5-flash` returned 404 for new projects. Fixed: default `gemini-flash-lite-latest`, model stays in Remote Config.

### AI NEEDS A PLAN B

App Check, quota and model changes all broke AI at some point; the hand-written path kept the flow alive.